

W. A. ACHIRA SADHARANGA

DATA SCIENCE UNDERGRADUATE | ASPIRING DATA ANALYST | PYTHON & POWER BI

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SUMMARY

Data Science undergraduate at Sri Lanka Technology Campus (SLTC) with a strong interest in data analysis, machine learning, and software development. Proficient in Python, SQL, Excel, and Power BI, with hands-on experience gained through academic and collaborative projects. Passionate about solving real-world problems using data and seeking an IT Industry Internship to enhance my technical and professional skills.

EDUCATION

Bachelor of Science (Hons) in Data Science Sri Lanka Technology Campus (SLTC)	2023 - Present
G.C.E. Advanced Level – Engineering Technology Richmond College, Galle.	2022

ACADEMIC & RESEARCH PROJECTS

End-to-End Explainable AI and 3D Visualization Framework for Vehicle Failure Management Final Year Research Project – Ongoing | Group Project

Developing an AI-based predictive maintenance framework for Toyota Aqua (Prius C) vehicles using machine learning, Explainable AI, and 3D visualization.

- Developed Random Forest and XGBoost models to predict engine overheating and low oil-pressure faults using OBD-II data.
- Applied SHAP and LIME to provide explainable fault analysis and transparent model predictions.
- Designed an interactive 3D engine visualization system for fault identification and maintenance recommendations.
- Evaluated system performance and usability against traditional vehicle warning systems.

Technologies: Python, Scikit-learn, Random Forest, XGBoost, SHAP, LIME, Flask, Three.js

Public Pulse – Grounded Multilingual Civic Intelligence & Graph Learning Platform | Group Project

Developed an AI-powered multilingual platform combining NLP, LLMs, RAG, and Graph Neural Networks to analyze Sri Lankan public discourse and academic citation networks.

- Built a 3-stage XLM-RoBERTa NLP pipeline for YouTube comment noise filtering, five-domain policy classification, and stance detection.
- Developed a RAG pipeline with Google Gemini to generate grounded, citation-backed public sentiment summaries.
- Implemented an NLI-based Faithfulness Auditor to verify AI-generated claims against supporting evidence.
- Developed a GCN/GAT-based Graph Neural Network for classifying 169K+ academic papers using the OGBN-Arxiv citation network.
- Achieved 58.6% accuracy with GCN, with 3.25× faster training compared with GAT.
- Applied PCA and t-SNE for model representation analysis and visualization.
- Built interactive React + TypeScript and Streamlit dashboards for analytics, model results, heatmaps, and trend visualizations.
- Achieved 278 passing tests across the end-to-end system.

Technologies: Python, PyTorch, XLM-RoBERTa, NLP, LLMs, RAG, Google Gemini, Hugging Face, GNN, GCN, GAT, Machine Learning, PCA, t-SNE, React, TypeScript, Streamlit, YouTube Data API

Traffic Rule Violation Detection System Using Image Processing | Group Project

Developing an AI-powered traffic monitoring system that automatically detects road violations from live camera feeds and recorded video.

- Leading machine learning development, research, model training, optimization, and ANPR implementation.
- Developing YOLO-based computer vision models to detect red-light violations, illegal parking, speeding, and other traffic violations.
- Implementing Automatic Number Plate Recognition (ANPR) using a custom CNN-based license plate recognition approach.
- Designing an evidence management system to store violation images, timestamps, locations, and license plate information.
- Integrating ML models with a web-based dashboard for traffic violation management.

Technologies: Python, OpenCV, YOLO, TensorFlow/PyTorch, CNN, ANPR, PostgreSQL, Django/Flask

Vow & Vision – AI-Powered Wedding Planning Platform | Group Project

- Designed and developed an AI-assisted wedding planning platform to streamline vendor management, budgeting, and guest coordination.
- Created user-centred UI/UX designs in Figma.
- Achieved a 100% success rate in usability testing using Maze.
- Collaborated with team members to improve user experience through human-centred design principles.

Technologies: Figma, Maze, Human-Centered Design, UI/UX Design

TECHNICAL SKILLS

- Programming: Python, Java, JavaScript, SQL
- Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Excel, Power BI
- Machine Learning & AI: Scikit-learn, PyTorch, XGBoost, Random Forest, XLM-RoBERTa, NLP, LLMs, RAG, SHAP, LIME
- Big Data & Data Engineering: PySpark, Apache Hadoop, Apache Airflow, MLflow
- Databases: MySQL, PostgreSQL, MongoDB
- Frameworks & Development: Flask, Flutter, React, TypeScript, HTML, CSS
- Tools: Git, GitHub, DAGsHub, Docker, Jupyter Notebook, Google Colab, VS Code, Streamlit, Jira

LEADERSHIP & VOLUNTEERING

- Founder, Information Technology & Industry Association (ITIA), Faculty of Computing & IT, SLTC.
- Club President, Leo Club of SLTC. (2025-2026)
- Faculty Representative, Faculty of Computing & IT, SLTC.
- Senior Moderator, Media Unit, SLTC.
- Committee Head, SLTC Student Interactive Society. (2024-2025)
- Member, IEEE Student Branch.
- Leadership Board Leader, AIESEC Colombo South. (2025-2026)

REFERENCES

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